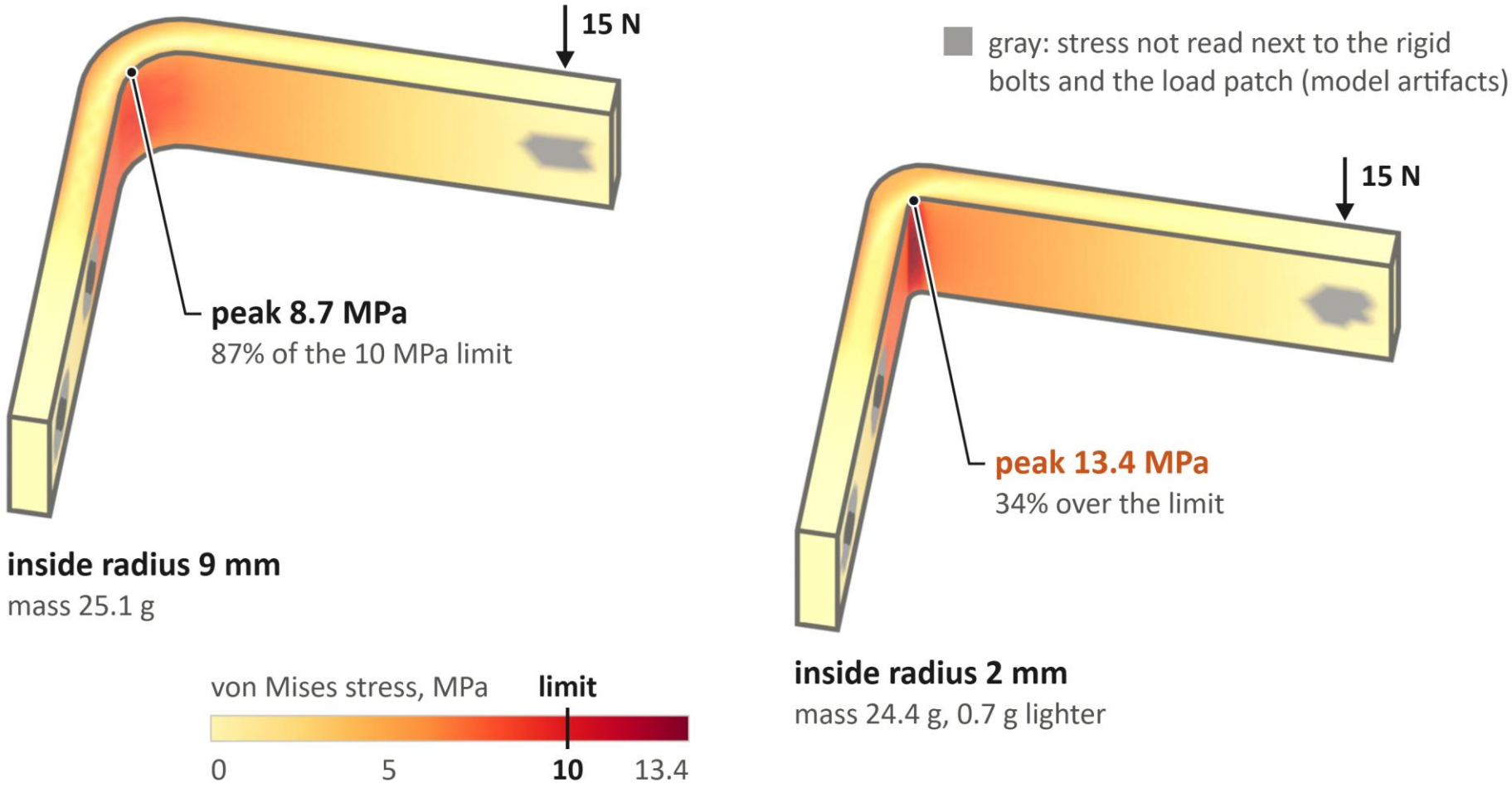
FEAther: A Self-Learning Multi-Agent System for Lightweight Load-Bearing Part Design with Finite Element Analysis

Capstone research proposal, Research Methods, Lab 8

Rifqi Wibisono · Nanjing Xiaozhuang University · November 2026

Tightening a bend from 9 mm to 2 mm inside radius raises its peak stress by half; the shape barely changes



FEA by the FEAther checker (CalculiX, quadratic tetrahedra) on reference parts L3 and L1: same 15 N load at 70 mm, same 6 × 24 mm section, same bolts. Stress is not read in the gray zones (model artifacts).

No work yet trains a model to choose a part's layout against physics

The model decides:	**judged by resemblance** to a reference shape	**judged by physics,** at run time only	**trained on physics** results
the whole program: layout and sizes	• **CAD-Coder [3]** text to CadQuery, RL on a shape reward • **cadrille [4]** point clouds, images, text; RL on a shape reward • **CAD-Recode [2]** point clouds to CadQuery	• **Son et al. [8]** frontier agents iterate on FEA reports, untrained • **CADEngBench [9]** benchmark, no training	• **FEAther** trains the model that writes the whole program on FEA results, rewarding low mass *no prior work in this cell*
only the numbers in a fixed template	*none found*	*none found*	• **COSMO-Agent [11]** trained with CalculiX, but only tunes 25 templates

Placement of each work from the paper itself; arXiv search of 1 November 2026. Numbers in brackets: references, last slide.

The designer is checked by physics, revises until the part passes, then lightens it while it still passes

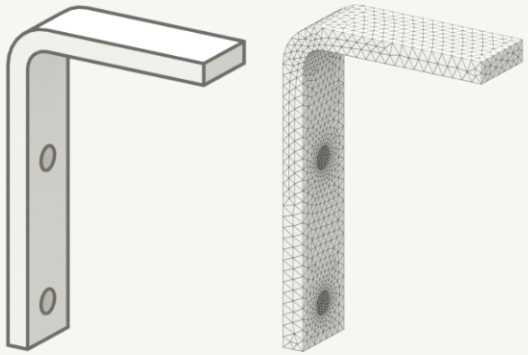
Spec
from the parser

10 N down at (42, 0, 76) mm
2 M5 bolts at z = 10, 43 mm
inside x 0–60, y ±15 mm
peak stress < 10 MPa
deflection < 3.8 mm
mass budget 9.3 g

Designer
the trained LLM writes CadQuery

```
t = 5  # thickness
r_i = 5  # bend radius
b = 16.5  # width
x_tip = 50
result = profile.extrude(
    b / 2, both=True)
for y, z, d in holes:
    result = result.cut(…)
```

Checker
runs it, meshes it, solves it



the part its FEA mesh

Report
back to the designer

PASS, reward 0.96
peak 7.77 MPa, limit 10
deflection 1.95 of 3.8 mm
mass 11.6 g, budget 9.3 g

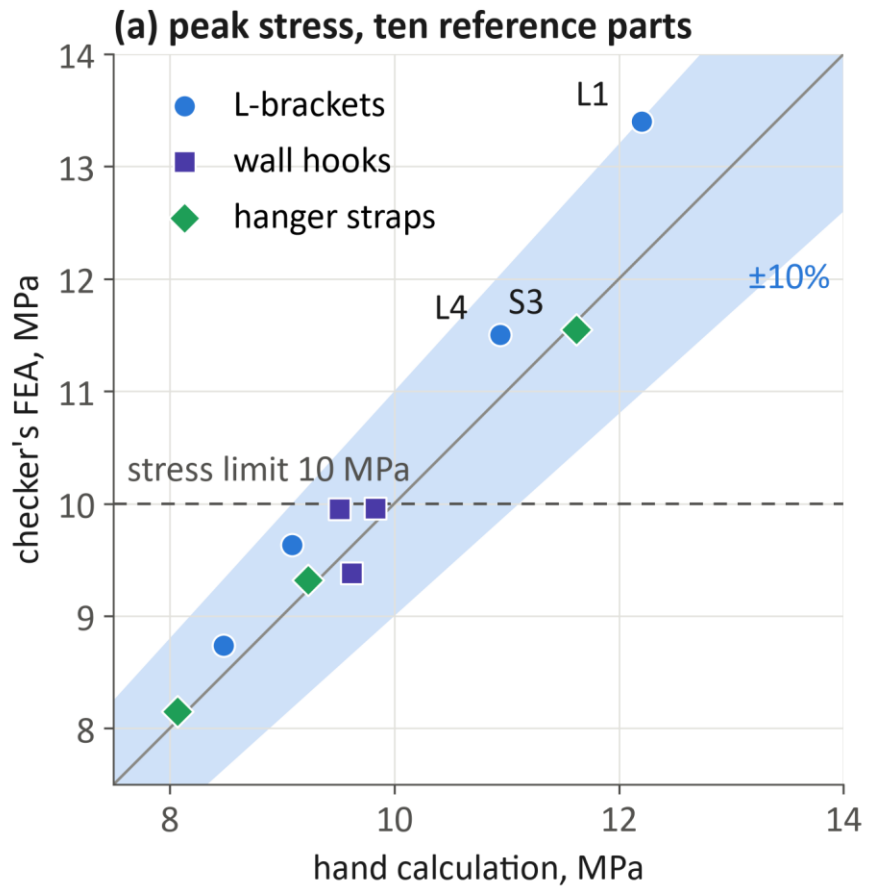
next request: lighten

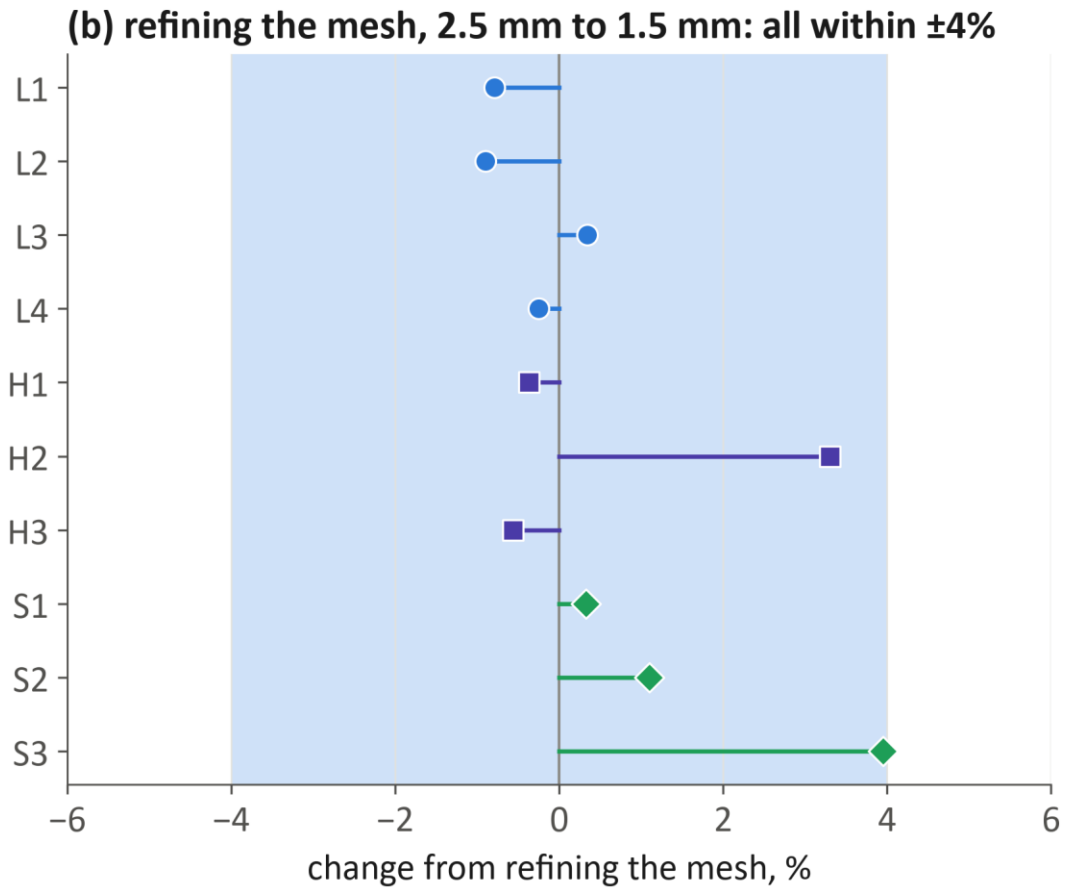
run time: revise while it fails, lighten while it passes

training: the reward (0.96 here) updates the designer's weights, by GRPO

One real pass: pilot spec l_bracket-0000, its reference program, the checker's mesh, and its report (FEAther repository).

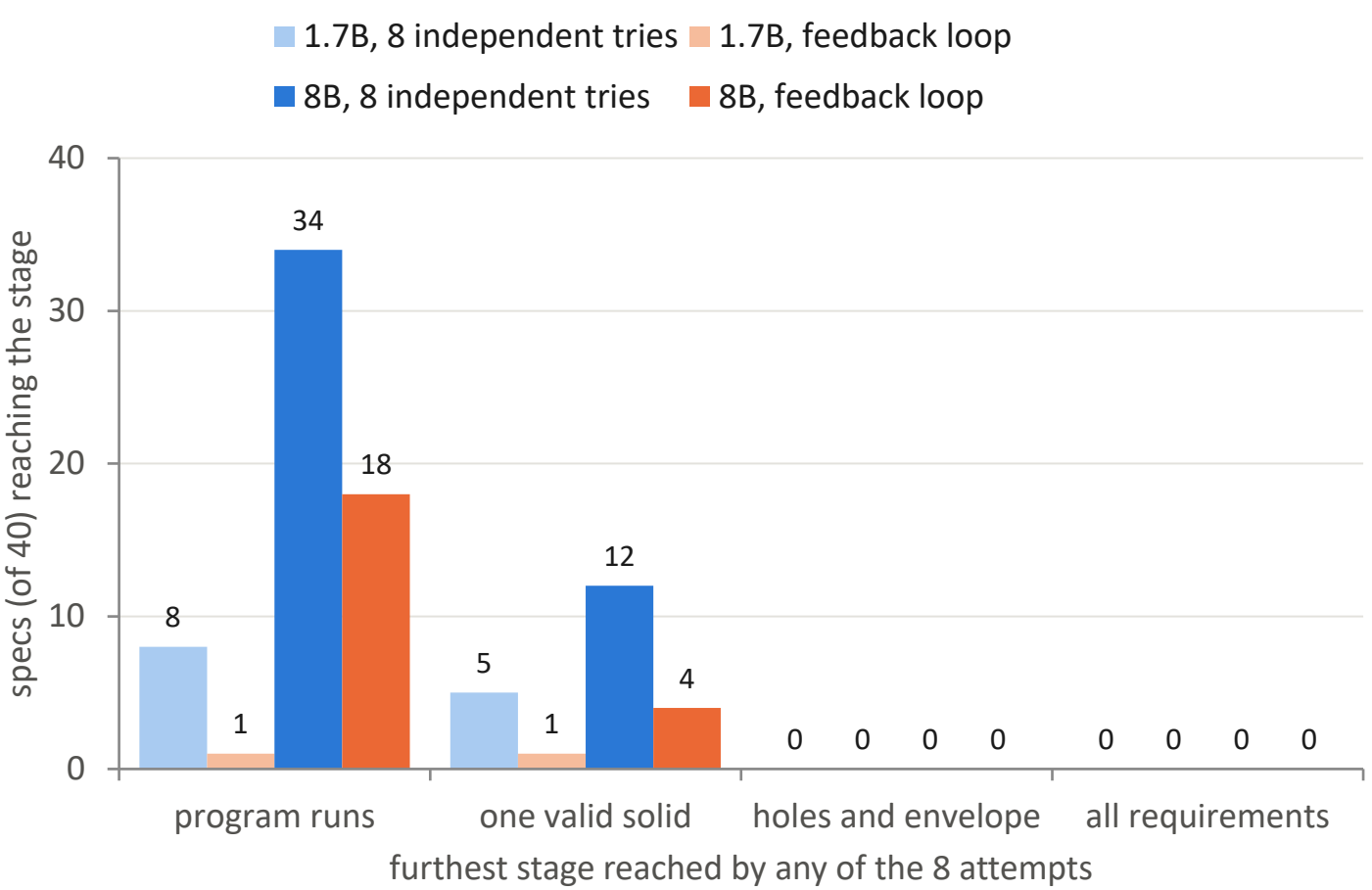
The checker's FEA agrees with textbook hand calculations within 10% on all ten reference parts

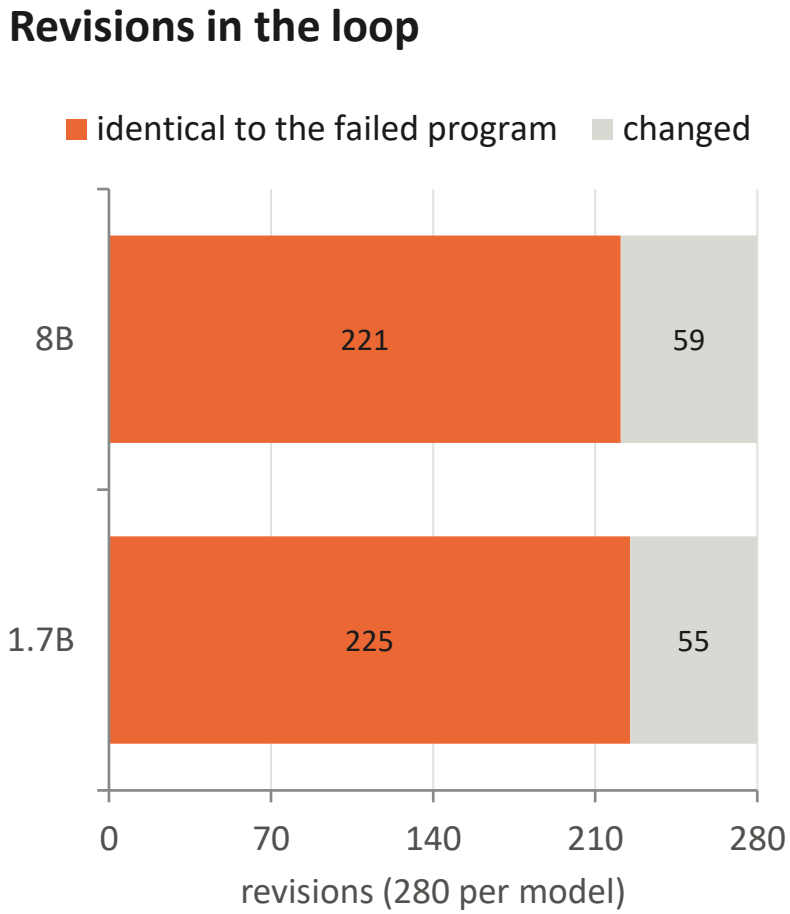




Hand calculations: curved-beam theory (Winkler) for bends, Heywood's formula for a hole in a strip [5], [6]. FEA: the checker, CalculiX with quadratic tetrahedra.

Untrained models pass none of the 40 specs, and 79% of their revisions repeat the failed program





FEAther baseline: Qwen3-1.7B (16-bit) and Qwen3-8B (FP8), thinking off, 8 checker runs per spec; 40 generated specs, each proven solvable.

Three questions test whether training on physics pays off

	Question	Hypothesis	Measured by	Test
RQ1	Does an FEA reward beat a shape reward?	It raises the full pass rate over SFT alone and over a shape reward	full pass rate on held-out specs and families	McNemar
RQ2	Does run-time feedback add anything beyond training, or only more attempts?	Untrained, independent tries beat the loop (pilot: 34 against 18 specs). Trained on revisions, the loop wins	specs passed within k = 1, 3, 5, 10 attempts; tokens per solved spec	McNemar at each k
RQ3	How much material do passing designs use?	With the mass term, passing designs are lighter than sizing alone	mass against sizing-only and topology-optimized references	Wilcoxon signed-rank

Hypotheses, baselines, and statistics: proposal, Sections III and VI.

References, numbered as in the proposal

[2] D. Rukhovich et al., “CAD-Recode: Reverse engineering CAD code from point clouds,” arXiv:2412.14042, 2024.

[3] Y. Guan et al., “CAD-Coder: Text-to-CAD generation with chain-of-thought and geometric reward,” NeurIPS 38, 2025.

[4] M. Kolodiazhnyi et al., “cadrille: Multi-modal CAD reconstruction with reinforcement learning,” ICLR, 2026.

[5] R. G. Budynas and J. K. Nisbett, Shigley’s Mechanical Engineering Design, 11th ed. McGraw-Hill, 2020.

[6] W. D. Pilkey and D. F. Pilkey, Peterson’s Stress Concentration Factors, 3rd ed. Wiley, 2008.

[8] G. Son et al., “Self-improving CAD generation agents with finite element analysis as feedback,” arXiv:2605.17448, 2026.

[9] H. Singh et al., “CADEngBench: It looks like CAD, but does it work?” arXiv:2608.09296, 2026.

[10] T. X. Olausson et al., “Is self-repair a silver bullet for code generation?” ICLR, 2024.

[11] L. Deng et al., “COSMO-Agent: Tool-augmented agent for closed-loop optimization, simulation, and modeling orchestration,” arXiv:2604.05547, 2026.